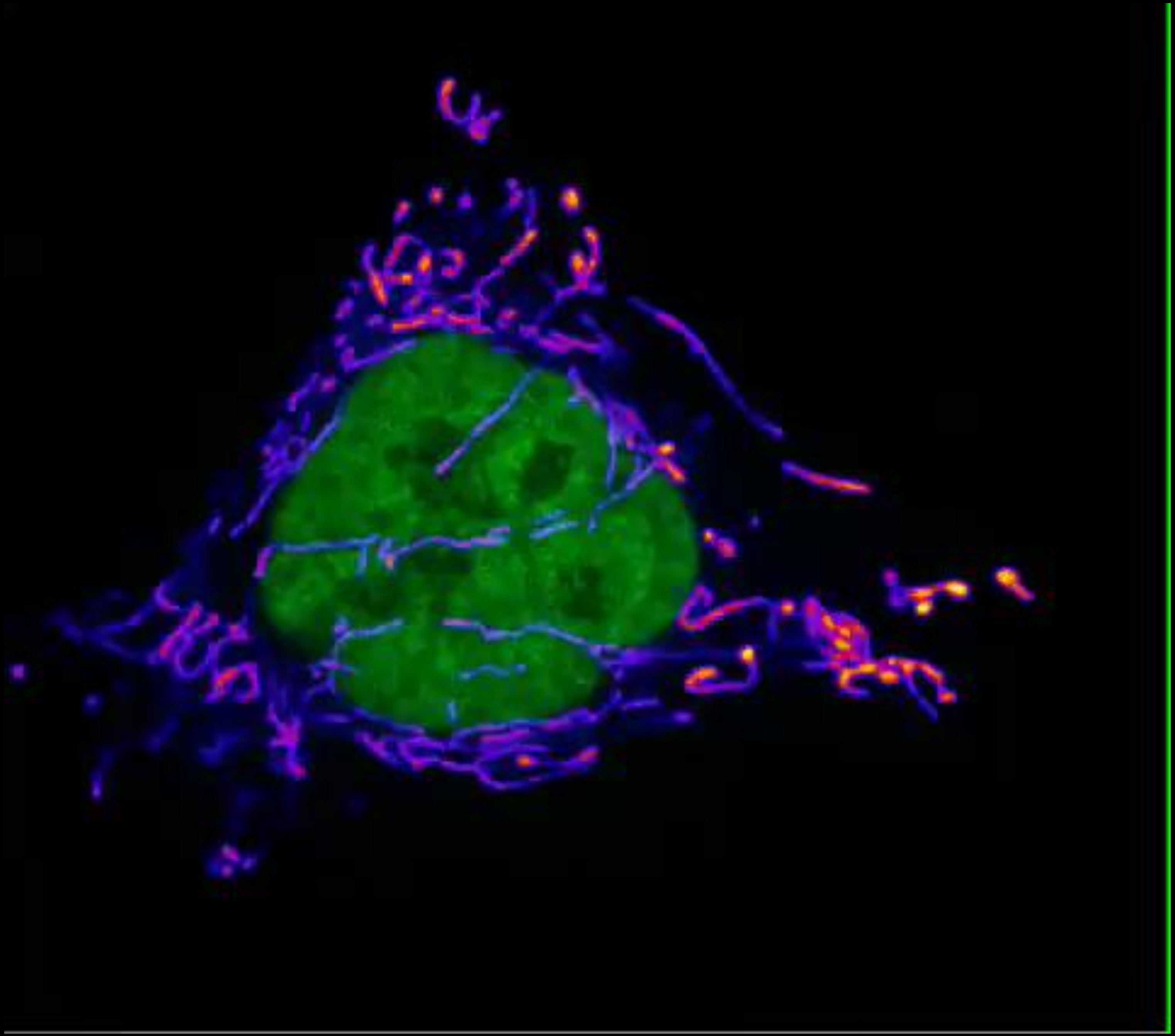

Basic Principles of Medicine 2
Module : Digestion and Metabolism
Lecture No: 08

Title: ETC and Oxidative Phosphorylation

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Electron transport chain and Oxidative Phosphorylation

SOM. MK.I.BPM2.2.DM.3.BCHM.1120	Identify clinical features of mitochondrial disorders
SOM. MK.I.BPM2.2.DM.3.BCHM.1121	List diseases involving mutations in mitochondrial DNA: Leber Hereditary Optic Neuropathy, Myoclonic Epilepsy and Ragged-Red Fiber Disease, MELAS
SOM. MK.I.BPM2.2.DM.3.BCHM.1122	Outline the aspartate-malate and glycerophosphate (primarily white muscle tissue) shuttles and their significance in regenerating cytoplasmic NAD+
SOM. MK.I.BPM2.2.DM.3.BCHM.1123	Outline the different types of electron carriers found in the ETC complexes and follow the path electrons from NADH and FADH2 to oxygen
SOM. MK.I.BPM2.2.DM.3.BCHM.1124	Discuss the mode of action/effects of specific inhibitors of the ETC (including antimycin A, azide, hydrogen sulphide, cyanide, rotenone, piericidin A, carbon monoxide)
SOM. MK.I.BPM2.2.DM.3.BCHM.1125	Differentiate between: direct inhibitors of the ETC, uncouplers of oxidative phosphorylation, ATP synthase inhibitors (oligomycin) and inhibitors of the ATP/ADP translocase
SOM. MK.I.BPM2.2.DM.3.BCHM.1126	Explain how the electrochemical gradient is generated during electron transport to drive ATP synthesis and consider Mitchell's Chemiosmotic theory
SOM. MK.I.BPM2.2.DM.3.BCHM.1127	Describe the mode of action of mitochondrial ATP synthase and its inhibition by oligomycin
SOM. MK.I.BPM2.2.DM.3.BCHM.1128	Define uncoupling of oxidative phosphorylation, natural uncoupling in brown adipose tissue versus other types of uncouplers
SOM. MK.I.BPM2.2.DM.3.BCHM.1129	Discuss ADP-ATP translocase and inhibitors of the transporter (atractyloside and Bongkrekic acid)
SOM. MK.I.BPM2.2.DM.3.BCHM.1130	Discuss respiratory control of the ETC and the P/O ratio
SOM. MK.I.BPM2.2.DM.3.BCHM.1131	Identify the role of oxidative phosphorylation at conditions of low oxygen

Recommended Reading

- Lippincott’s Biochemistry 9th Edition
  🔗 https://sgu.idm.oclc.org/login?url=https://premiumbasicsciences.lwwhealthlibrary.com/book.aspx?bookid=3402
- Lippincott’s Illustrated Reviews in Biochemistry,9th ed Chapter 6, pages 80-89 Starting at “Electron Transport Chain” and “Phosphorylation of ADP to ATP” and summary
  - 6. Bioenergetics and Oxidative Phosphorylation | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library
  - See questions 6.1-6.4

Importance of ATP as an Energy Carrier

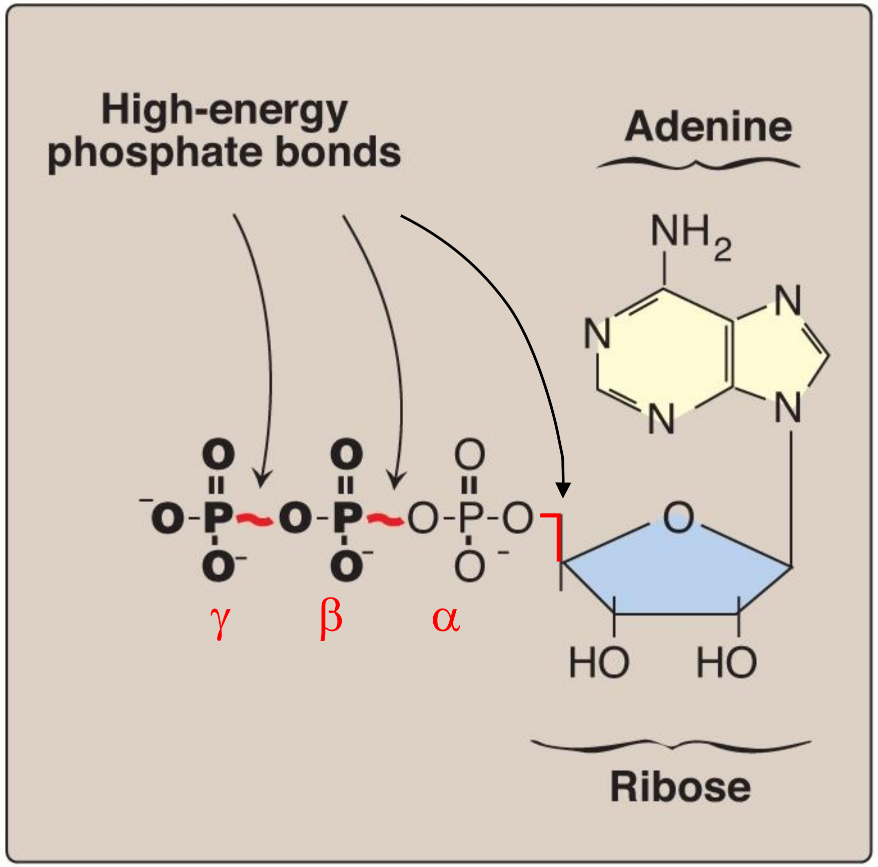


Fig 6.6

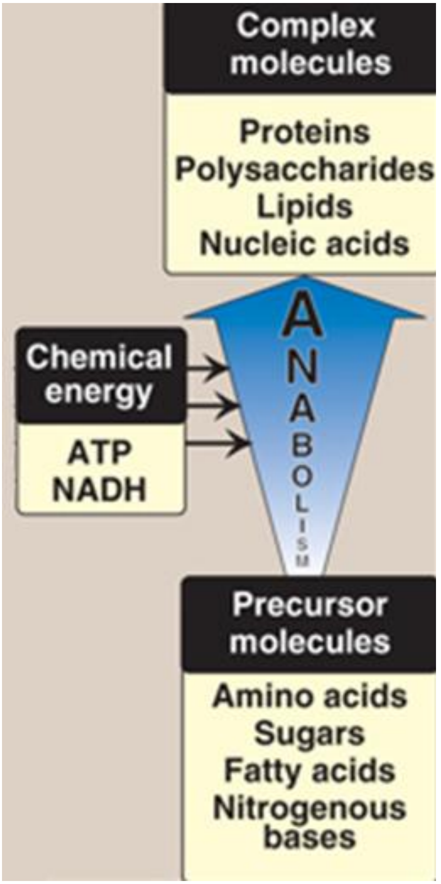
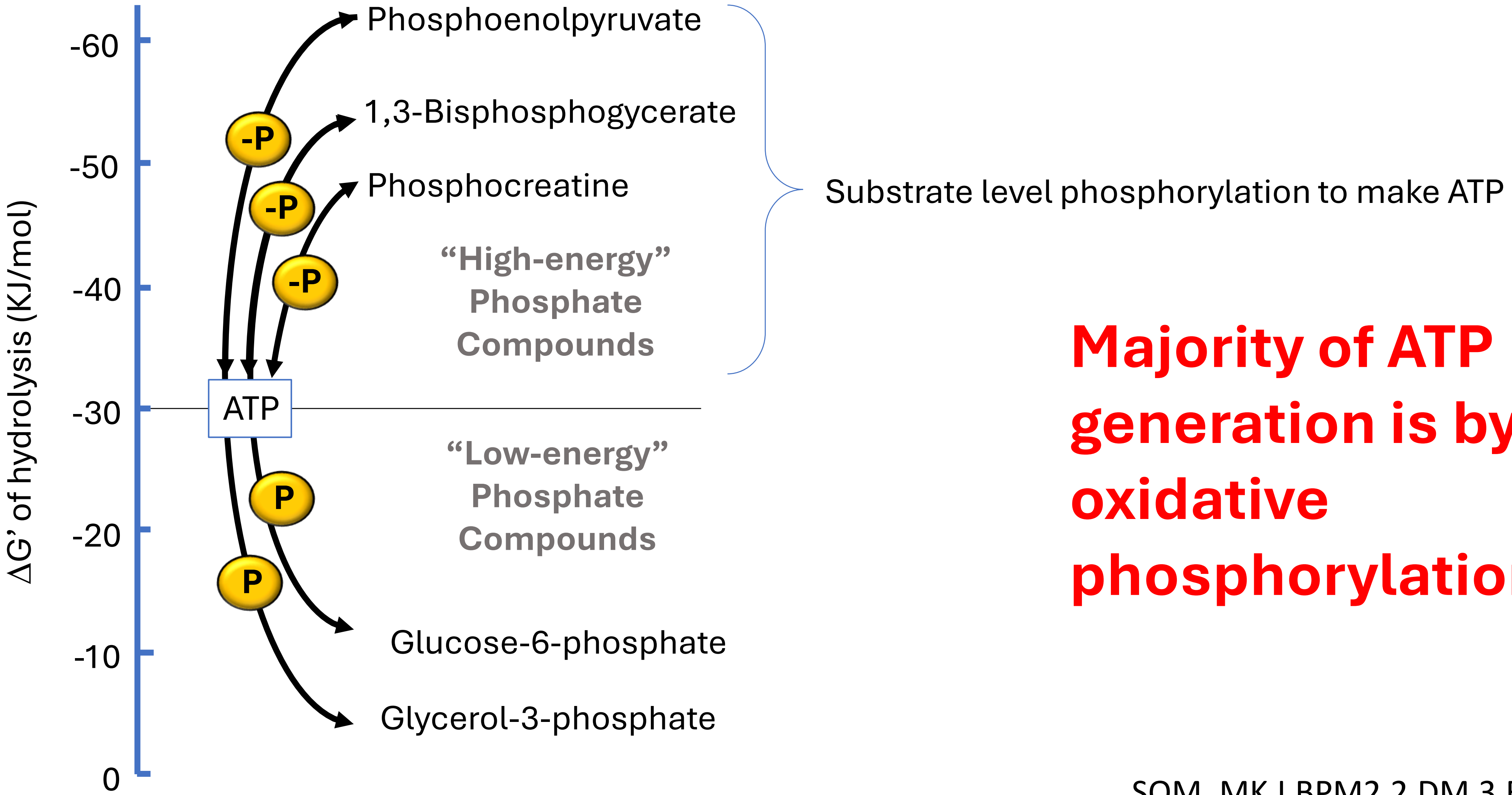


Fig. 8.4

High Energy Phosphate compounds can drive the formation of ATP from ADP

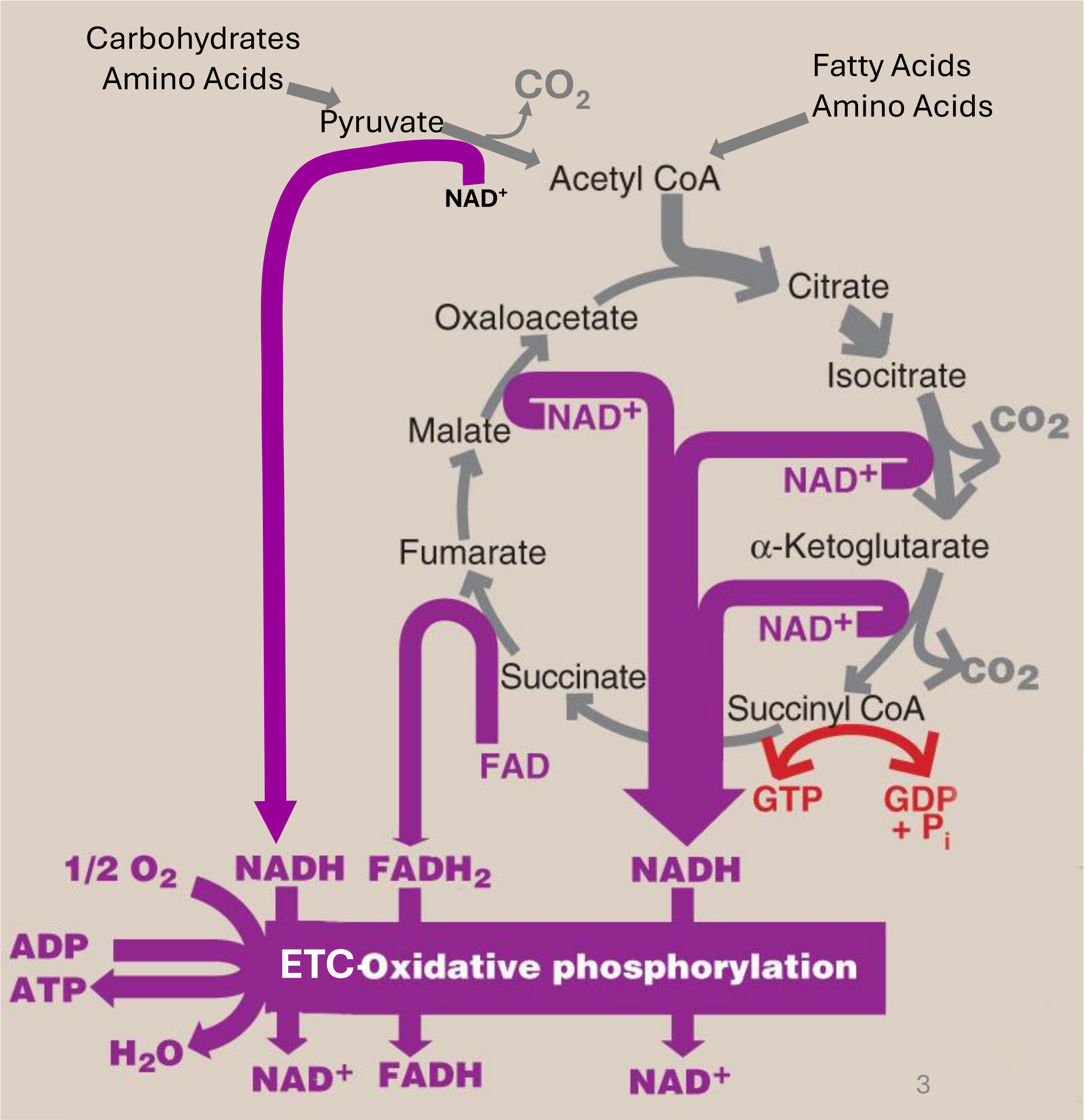


Majority of ATP generation is by oxidative phosphorylation

SOM. MK.I.BPM2.2.DM.3.BCHM.1130

Making ATP by Oxidative Phosphorylation

- Carbohydrates, fatty acids and amino acids are **oxidized** to CO_2 & H_2O
- Intermediates of reactions donate electrons to form REDUCED coenzymes NADH, $FADH_2$
- NADH and $FADH_2$ donate electrons to **Electron Transport Chain (ETC)**
- As electrons are passed down the ETC, they lose free energy
- This energy used to move protons across inner mitochondrial membrane and create a H^+ gradient
- The H^+ gradient drives oxidative phosphorylation: ATP synthesis
- Movement of electrons through ETC ultimately leads to **phosphorylation** of ADP to ATP "Oxidative Phosphorylation"

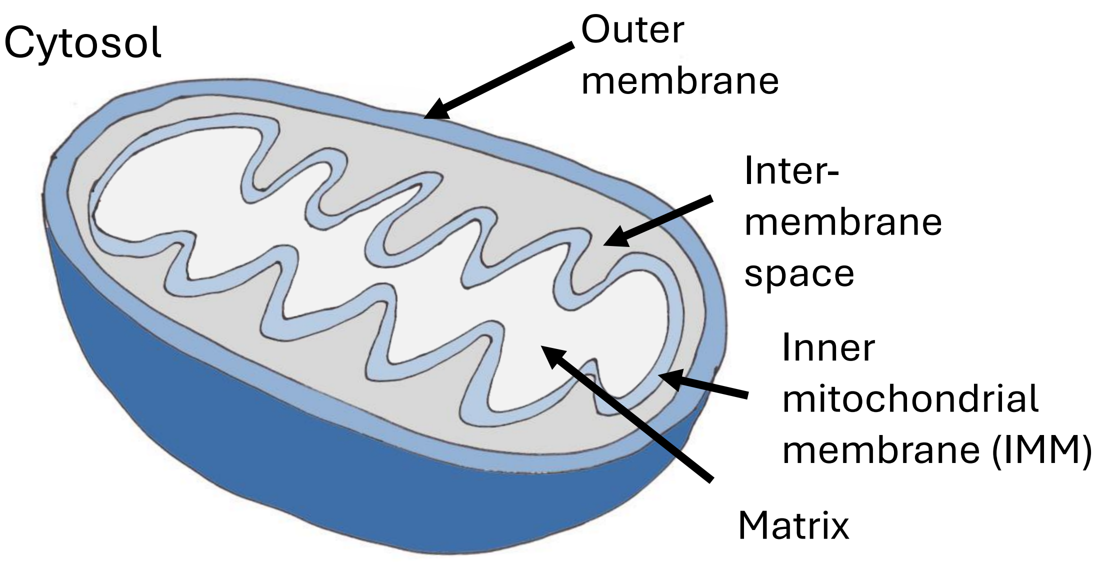


Adapted from fig 9.9

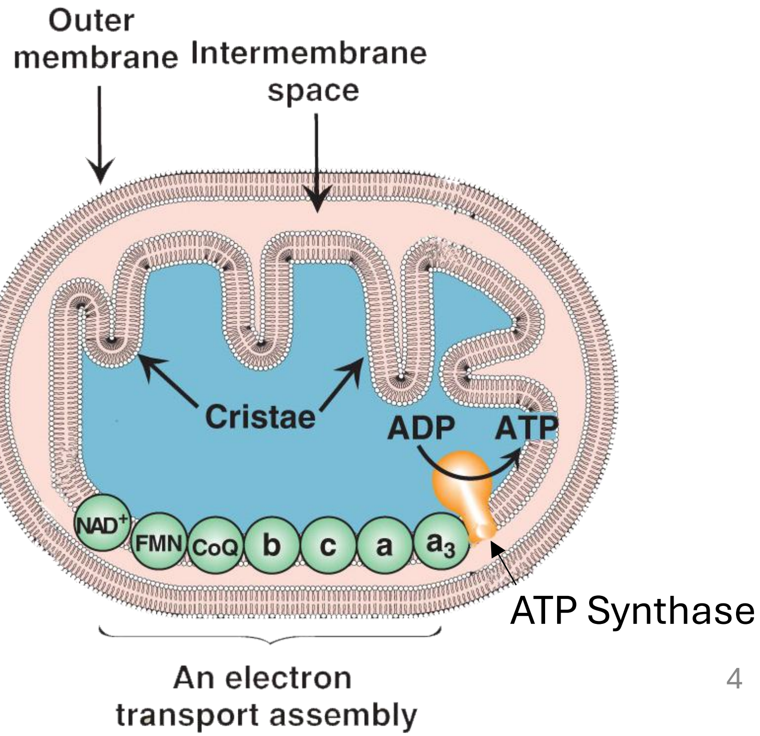
ATP generated by Oxidative Phosphorylation (OxPhos) (ETC coupled to ATP Synthase)

- Mitochondria – double membrane organelles
- Energy powerhouses
- Matrix – TCA Cycle enzymes
- Outer membrane permeable to most molecules
- Inner membrane highly impermeable
- Highly folded into invaginations - cristae.

- Enzymes of ETC and ATP synthase found on the Inner mitochondrial membrane.
- Inner membrane cristae increase membrane surface area and
- Inner membrane impermeability allows establishment of chemical gradients.

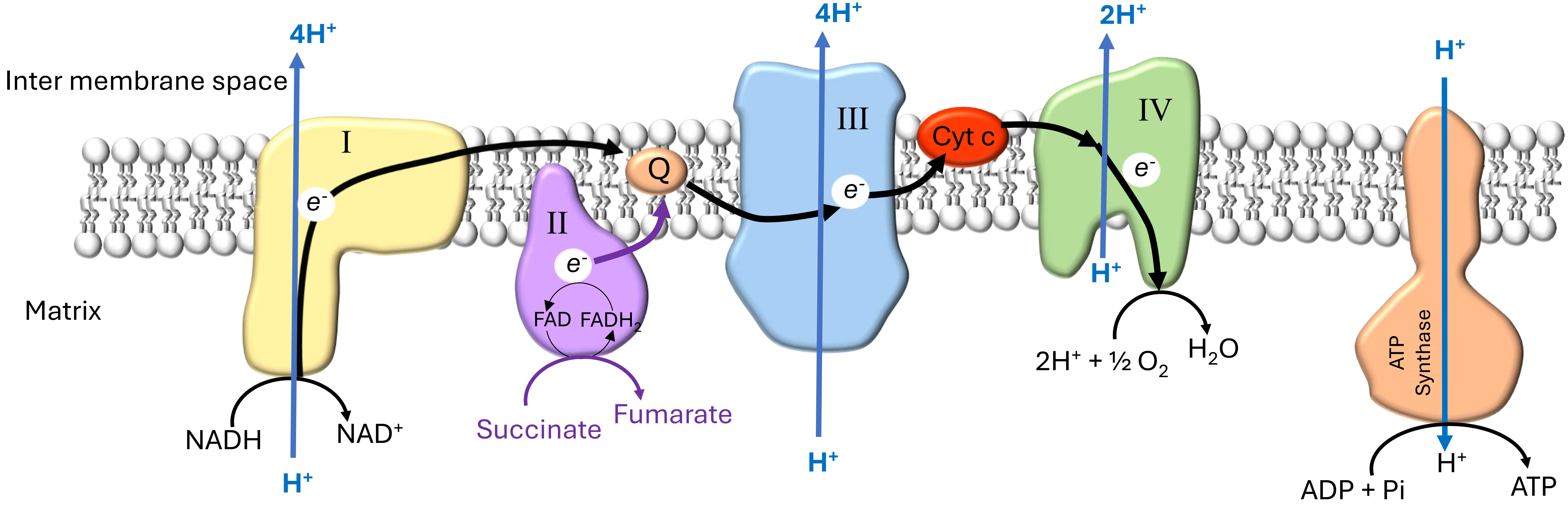


Mitochondrial Structure



ETC & OxPhos Function:

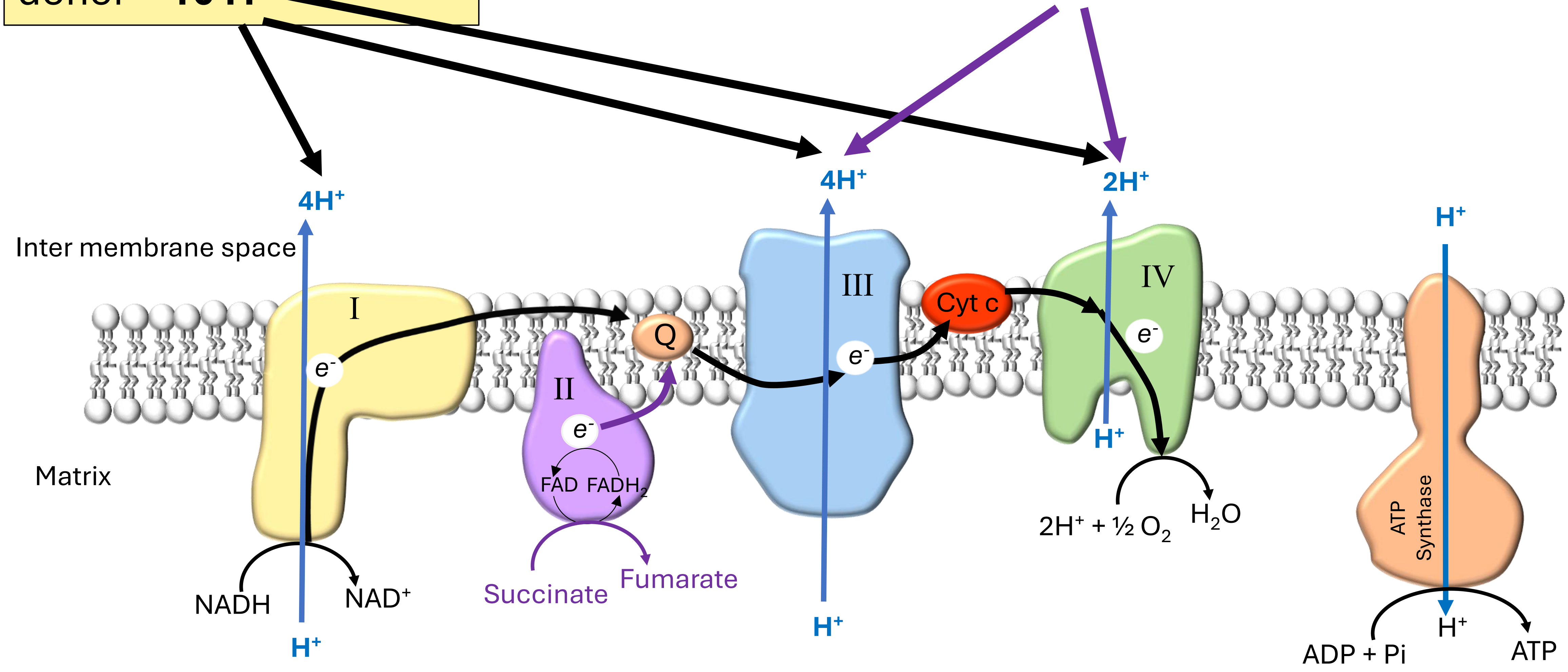
- Oxidize NADH and $FADH_2$
- Generate electrical energy by passing electrons to **Oxygen**
- Create a **proton gradient** across inner mitochondrial membrane
- Proton gradient drives phosphorylation of ADP to ATP
- **2 stages**
  - Electron transport then Oxidative phosphorylation



SOM. MK.I.BPM2.2.DM.3.BCHM.1124, 3 & 6

Protons moved against concentration gradient when **NADH** is electron donor = **10 H^+**

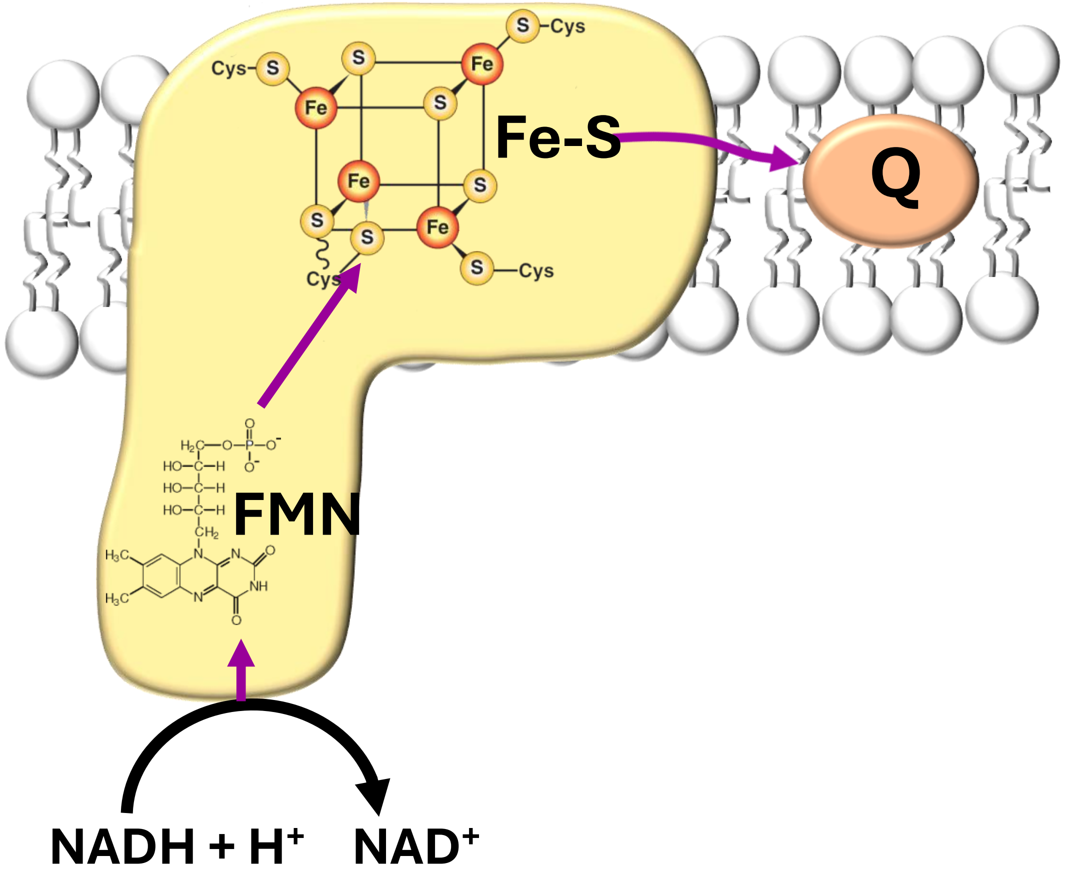
Protons moved against concentration gradient when **$FADH_2$** is electron donor = **6 H^+**



Complex I: NADH Dehydrogenase

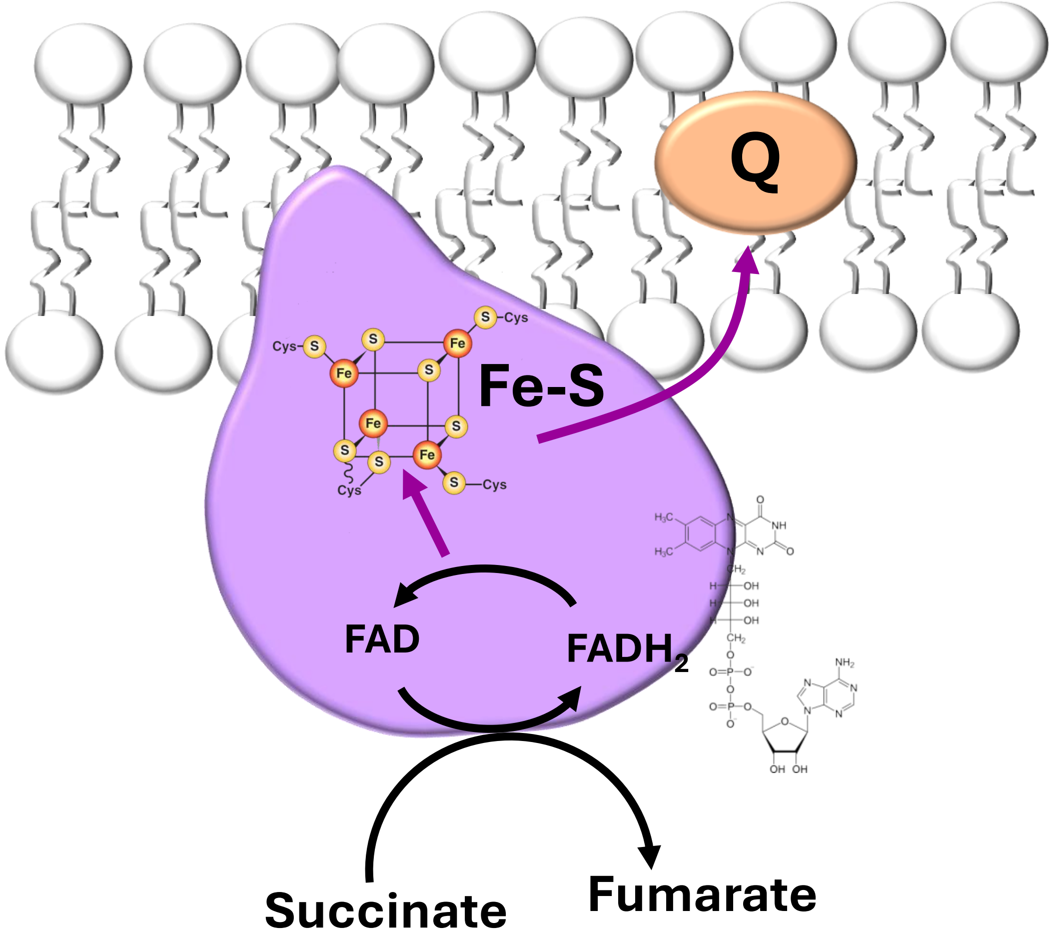
Amytal: barbiturate
Rotenone: insecticide
Piericidin A: bacterial antibiotic
Does not affect flow from complex II

- Oxidizes NADH & reduces Coenzyme Q (CoQ)
- Contains reversible electron acceptors
  - **FMN** derived from B2 (riboflavin)
  - Iron-sulfur (Fe-S) clusters
- NADH from TCA and PDH, fatty acid oxidation & glycolysis (via shuttles)
- Movement of electrons shown in magenta, from NADH to CoQ
  - Energy used to **pump protons** across inner-mitochondrial membrane to intermembrane space



Complex II: Succinate-Q-Reductase

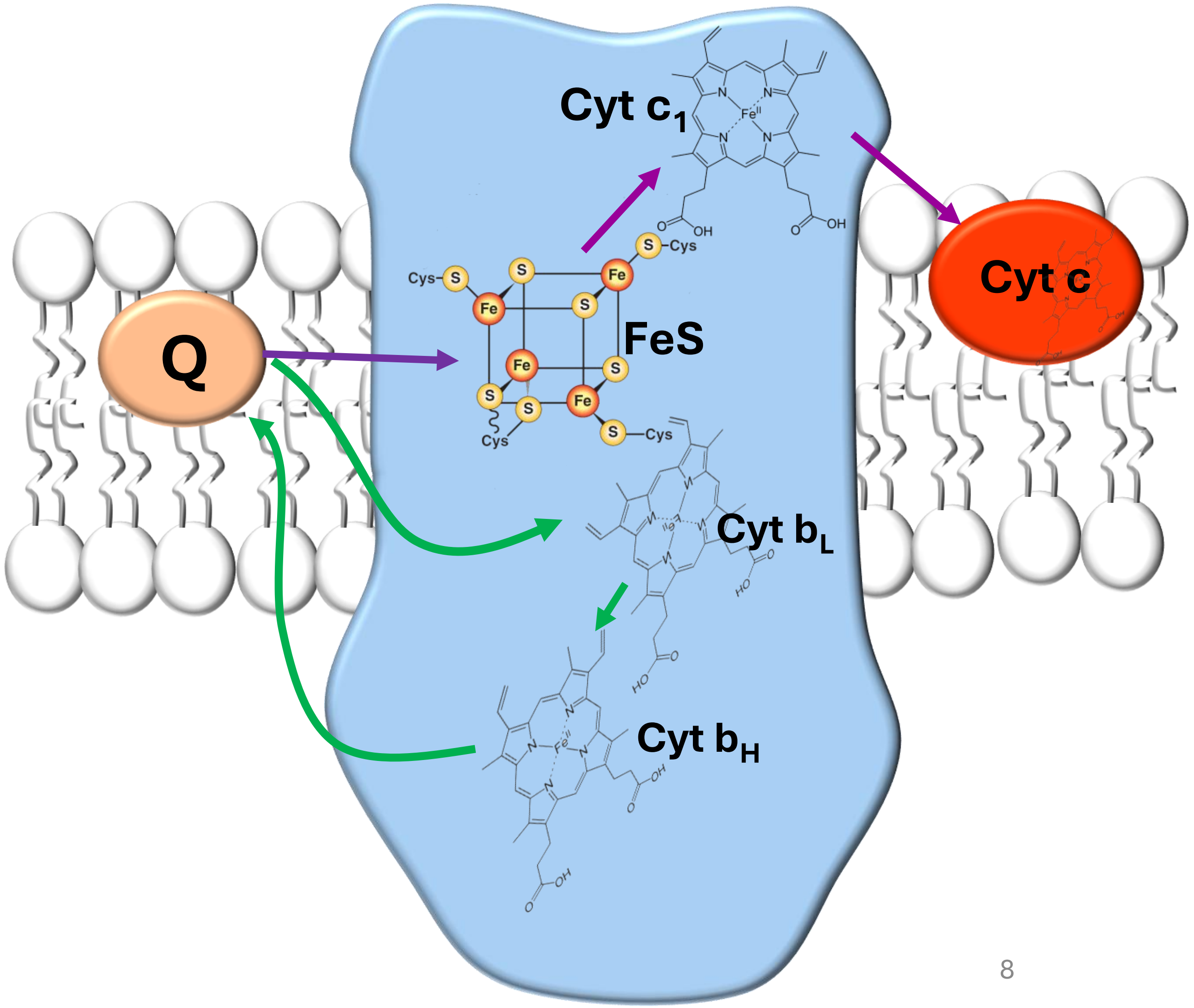
- Succinate dehydrogenase (TCA Cycle)
- Oxidizes succinate and reduces CoQ
- Electrons from succinate and FAO or glycerol phosphate shuttle ($FADH_2$)
- Contains
  - FAD derived from B2 (riboflavin)
  - Iron-sulfur (Fe-S)clusters
- Flow of electrons shown in magenta
- No protons are translocated



FAO=Fatty acid oxidation

Complex III-Cytochrome b-c_1 complex

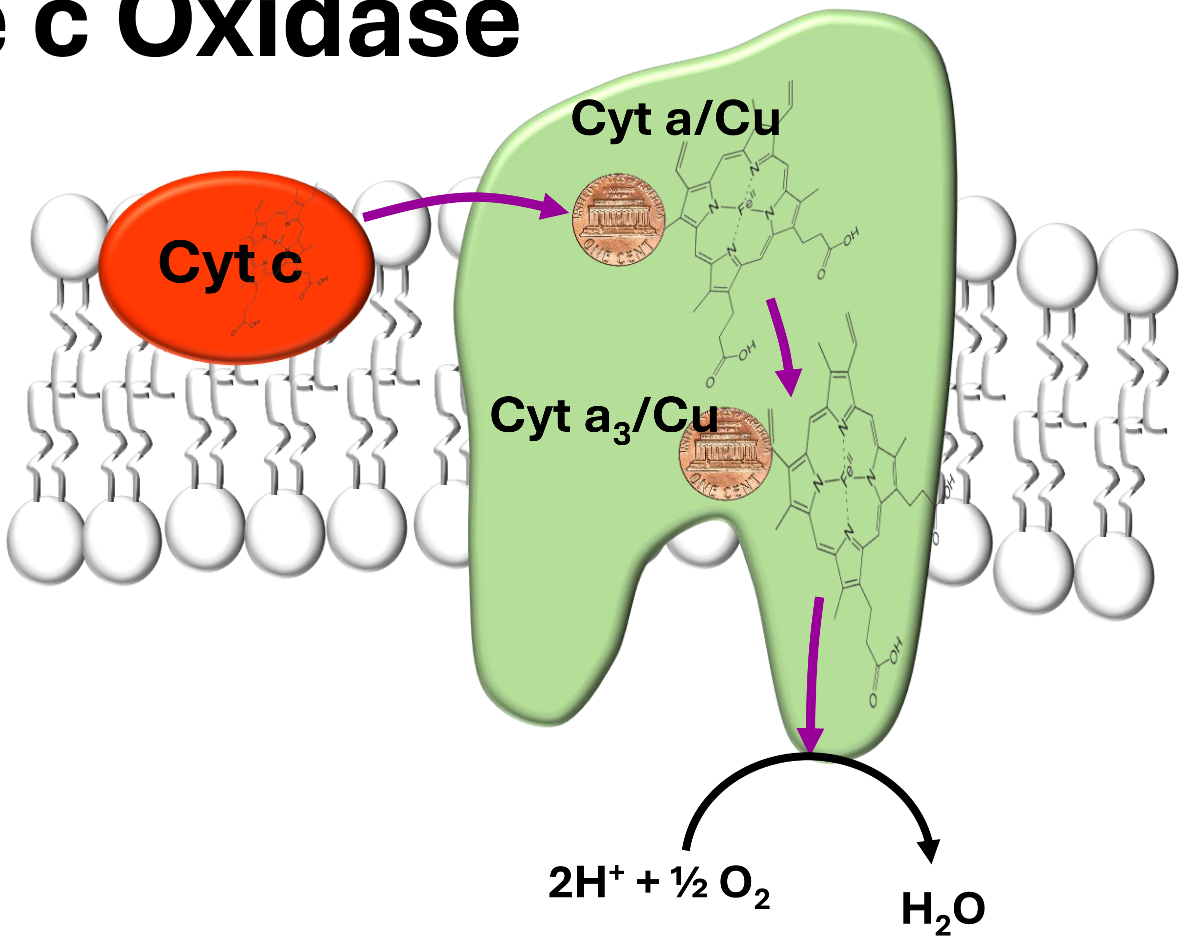
- Oxidizes CoQ and reduces Cyt c
- Movement of electrons shown in magenta, from CoQ to Cyt c
    - Energy used to **pump protons** across inner-mitochondrial membrane to intermembrane space
- Cytochrome b helps to regenerate CoQ (green)



Antimycin A: antibiotic
Inhibits cyt b

Complex IV: Cytochrome c Oxidase

- Oxidizes cytochrome c and reduces oxygen to water
- O_2 is final electron acceptor
- $O_2 \rightarrow H_2O$
- Contains heme groups with a bound copper atom
- Movement of electrons shown in magenta, from Cyt c to oxygen
  - Energy used to **pump protons** across the inner-mitochondrial membrane to the intermembrane space



Inhibitors of heme a_3-Cu

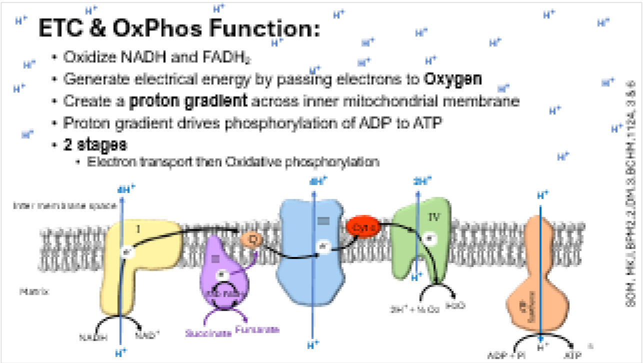
Cyanide: CN^- (bitter almond breath odor)

Carbon monoxide: CO (bright red color)

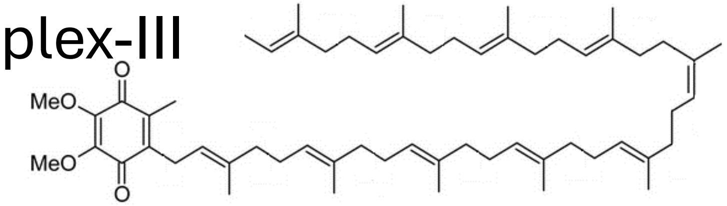
Azide: NaN_3

Hydrogen sulfide: H_2S

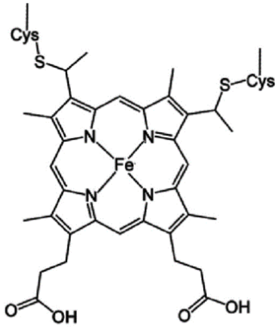
Mobile electron carriers



- **Coenzyme Q** or Ubiquinone or Q_{10} (ubiquitous "expression") and has a long hydrophobic tail
  - Lipid soluble molecule; Synthesized from farnesyl pyrophosphate (Cholesterol synthesis)
  - Transfers electrons from complex-I and complex-II to complex-III
  - **Statins** inhibit CoQ synthesis → Muscle pain



- **Cytochrome c:**
  - bound to intermembrane space side of the inner membrane
  - Contains Heme group
  - Acts as an electron shuttle between Complex III & IV
  - Cytochrome c release → Activates caspases →**Apoptosis**



ATP Synthase

Oligomycin

- Multi-subunit enzyme: F_0 and F_1 portions
  - F_0 (inner mitochondrial membrane) contains the **proton channel**
  - F_1 (in mitochondrial matrix) contains ATP synthase activity (ATP synthesis)
- Protons pumped to intermembrane space re-enter matrix through the H+ channel in the F_0
- Proton reentry drives conformational changes (rotation of the C ring) which then drives ATP synthesis

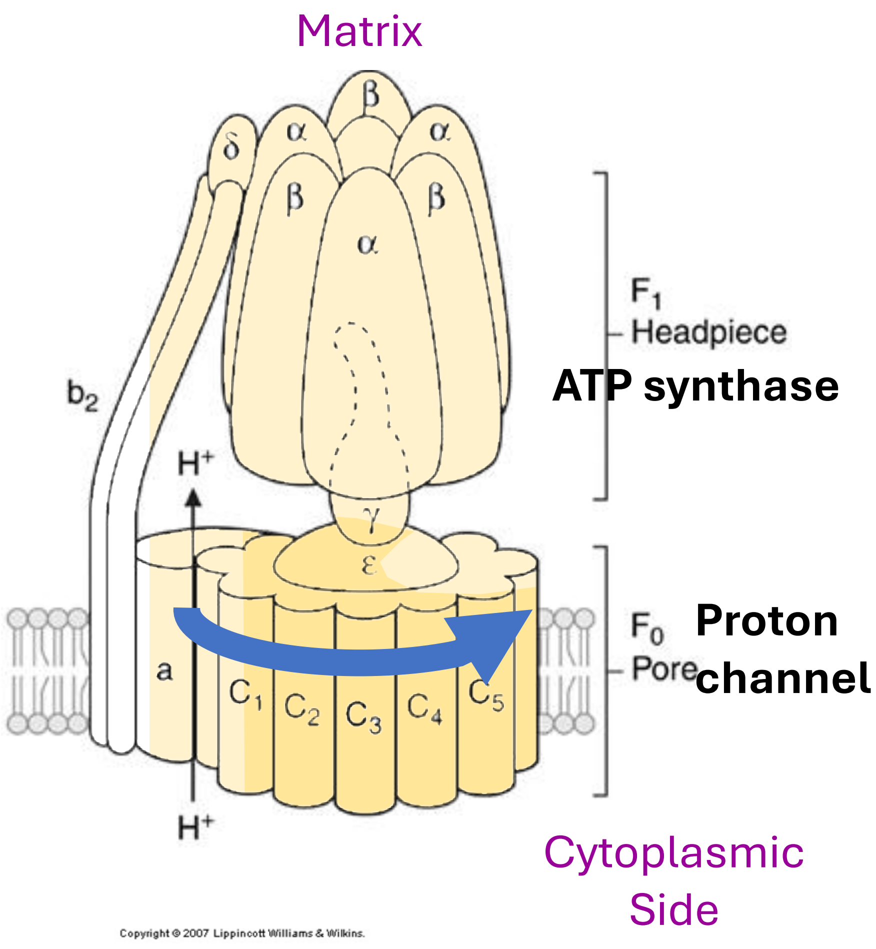
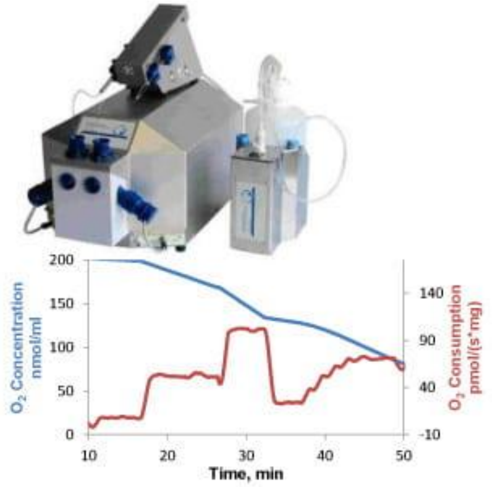


Fig 6.14

P/O Ratio (Phosphate to Oxygen Ratio)

- **Complex I and III** pump 4 H^+ & **Complex IV** pumps 2 H^+ into intermembrane space

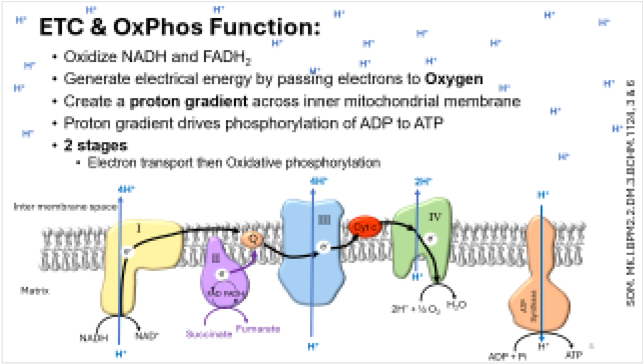
- **~3-4 H^+** required to synthesize 1 ATP
  - $\Rightarrow$ NADH entering at **Complex I** pumps 10 H^+ = 2.5 - 3 ATP
  - $\Rightarrow$ $FADH_2$ entering at **Complex II** pumps 6 H^+ = 1.5 - 2 ATP



- P/O ratio = Number of ATP synthesized per atom of oxygen used i.e. The number of ATP per ½ O_2
  - $\Rightarrow$P/O ratio for **NADH = ~3**
  - $\Rightarrow$P/O ratio for **$FADH_2$ = ~2**

Oxidative Phosphorylation & Chemiosmotic Theory: Proton Motive Force

Mitchell's Chemiosmotic Theory: 2 steps



Step 1

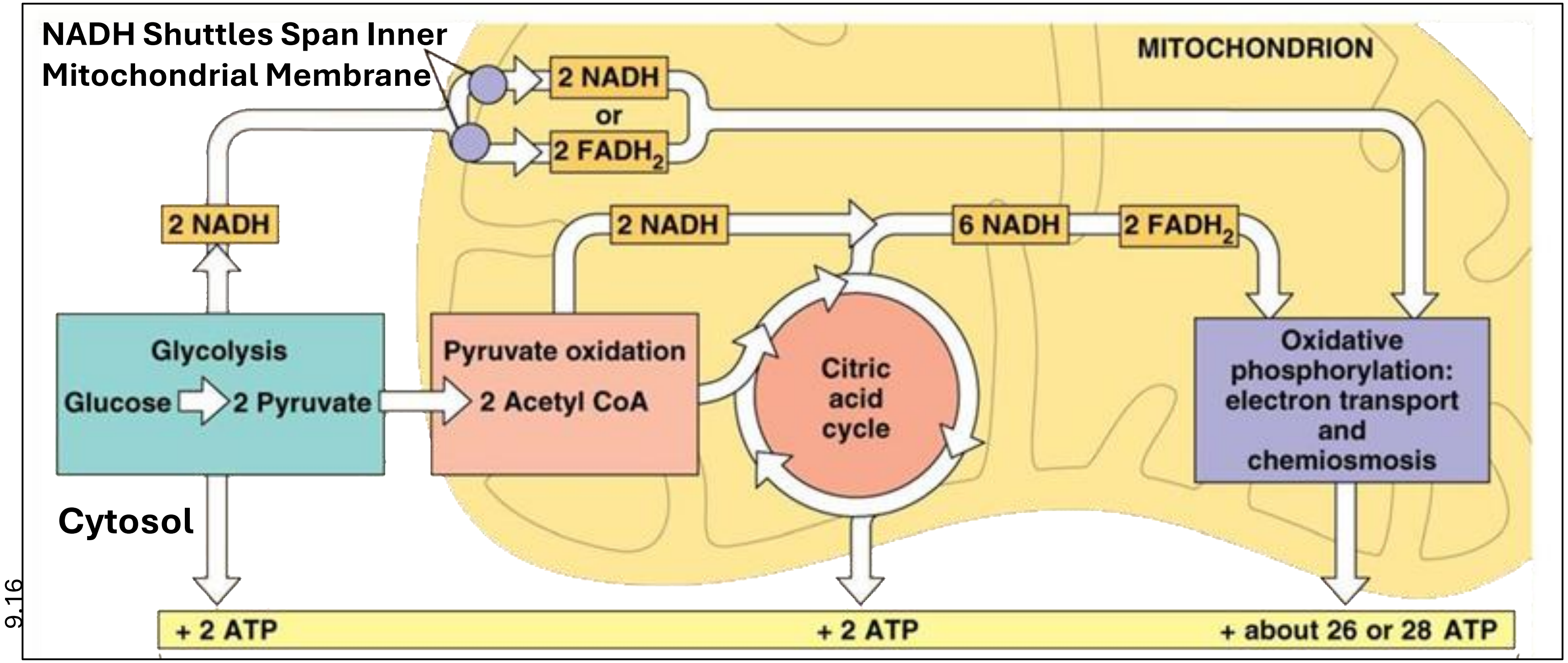
- As electrons flow down ETC, protons pumped into intermembrane space
- Protons pumped at complexes I, III & IV
- Protons cannot re-enter matrix → Creates a pH gradient

Step 2

- The pH gradient is used by **F_0F_1-ATP synthase (Complex V)**.
- Energy released by proton entry is coupled to ATP synthesis
- pH difference across the inner membrane ~0.75 pH units

ATP Generated Per Glucose Molecule

Adapted from Campbell Biology, 9th Ed. Pearson Fig. 9.16



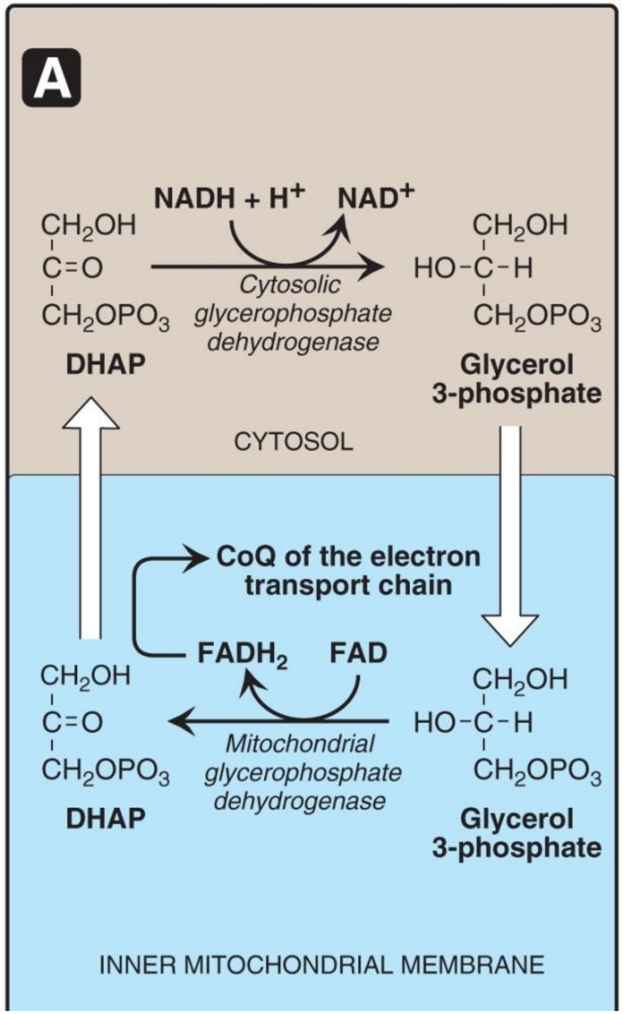
Glycolysis	Substrate level Phosphorylation	2 ATP
	2 NADH generation	4 or 6 ATP
Pyruvate Dehydrogenase	2 NADH	6 ATP
Citric Acid Cycle	Substrate level Phosphorylation	2 ATP
	6 NADH	18 ATP
	2 FADH2	4 ATP
	TOTAL	**36 or 38 ATP**

SOM. MK.I.BPM2.2.DM.3.BCHM.1123

NADH Shuttles

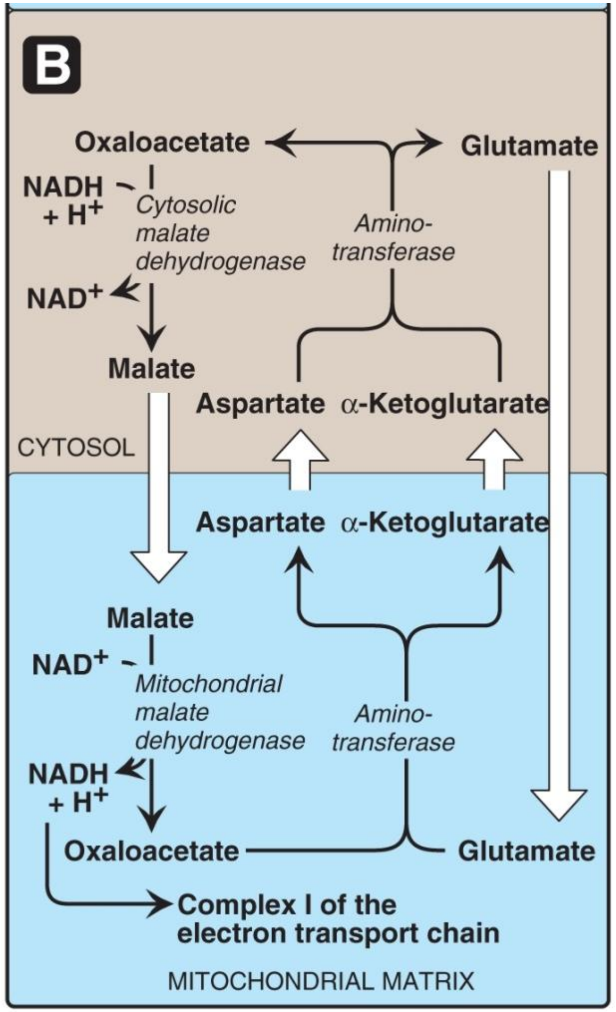
DHAP = dihydroxyacetone phosphate

Glycerophosphate Shuttle



Synthesis of 2 ATPs

Malate Dehydrogenase Shuttle



Synthesis of 3 ATPs

- **NADH (glycolysis)** cannot cross the Inner mitochondrial membrane
- Shuttles required to deliver NADH electrons from cytosol (glycolysis) to the mitochondrial matrix
- Glycerophosphate Shuttle = 2 ATP (forms $FADH_2$)
- Malate Dehydrogenase Shuttle = 3 ATP (Forms NADH)

Electron Transport will be active when:

[ADP]>[ATP] and [NADH]>[NAD^+]

Presence of oxygen

Hypoxia inhibits ETC and favors lactic acid formation

SOM. MK.I.BPM2.2.DM.3.BCHM.1127

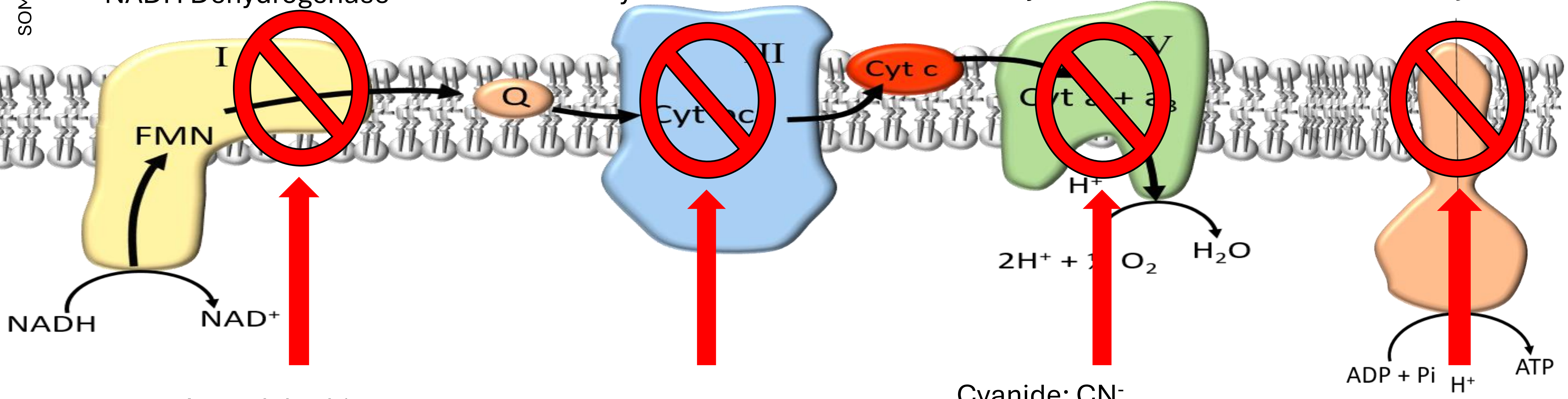
Inhibitors of ETC decrease: ATP synthesis, ETC activity & Oxygen consumption

Complex I
NADH Dehydrogenase

Complex III
Cytochrome reductase

Complex IV
Cytochrome c oxidase

Complex V
ATP Synthase



Amytal: barbiturate
Rotenone: insecticide
Piericidin A: bacterial antibiotic
Does not affect flow from complex II

Antimycin A: antibiotic
Inhibits cyt b

Cyanide: CN^-
Azide: NaN_3
Hydrogen sulfide: H_2S
Carbon monoxide: CO
All inhibit heme a_3-Cu

Oligomycin: antibiotic

Inhibitors of the ETC Complexes I, III & IV

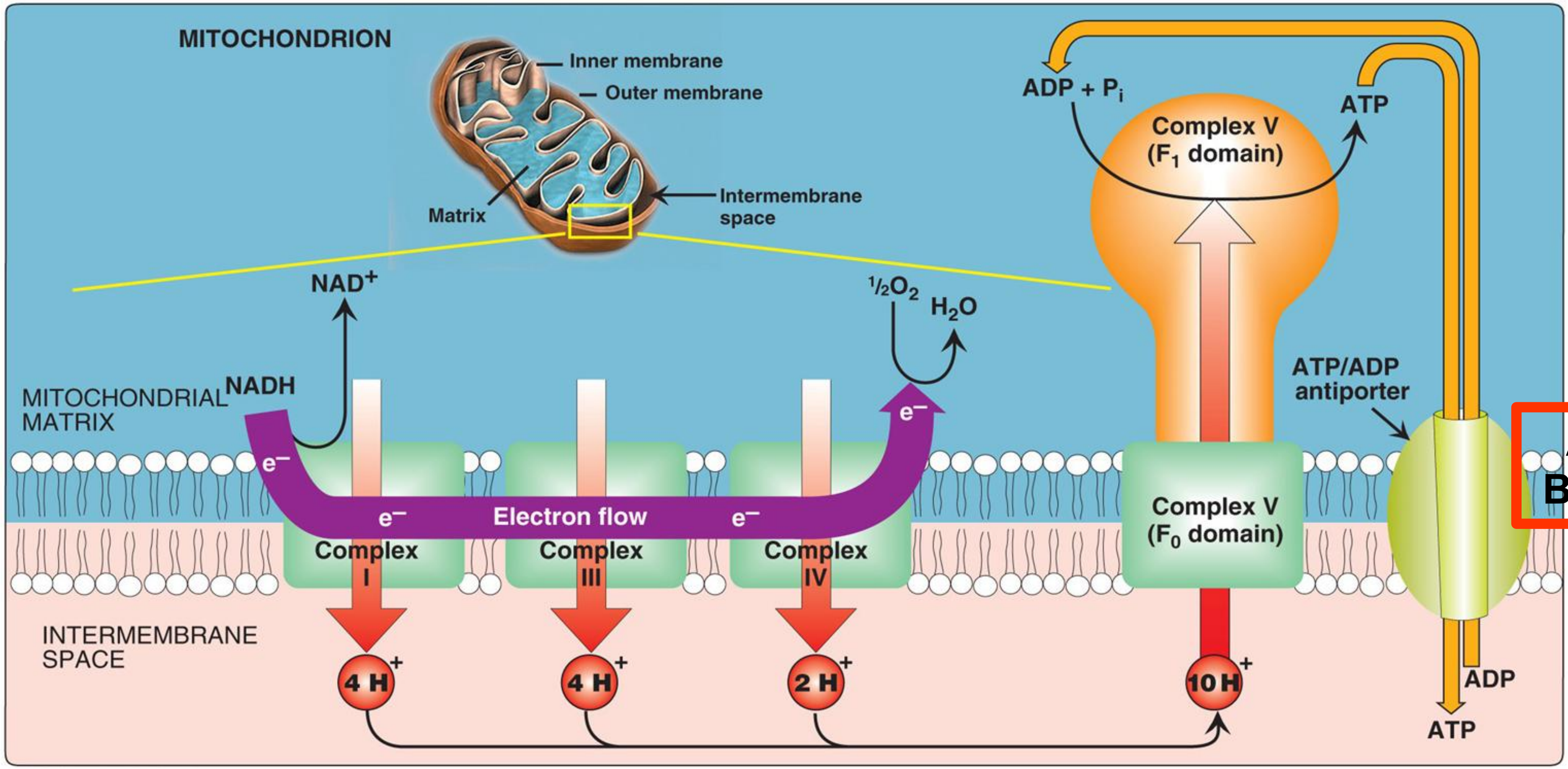
- Reduce electron transport and establishment of proton gradient
- ATP synthesis depends on the proton gradient, thus a reduction in electron transport results in a reduction of ATP synthesis

Inhibitors of ATP synthase: Oligomycin

- Electron transport is initially unaffected, and establishment of a proton gradient will occur
- Once a maximum gradient has been established, the **electron flow will be inhibited**

Antiport called Adenine Nucleotide Translocator (ANT)

ATP and ADP transported via an Antiport – Exchange, ATP out, ADP in

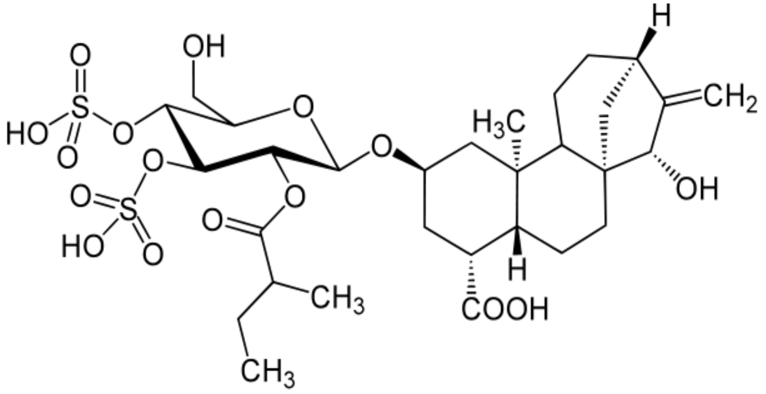


Atractyloside, Bongkrekic acid

Fig 6.13

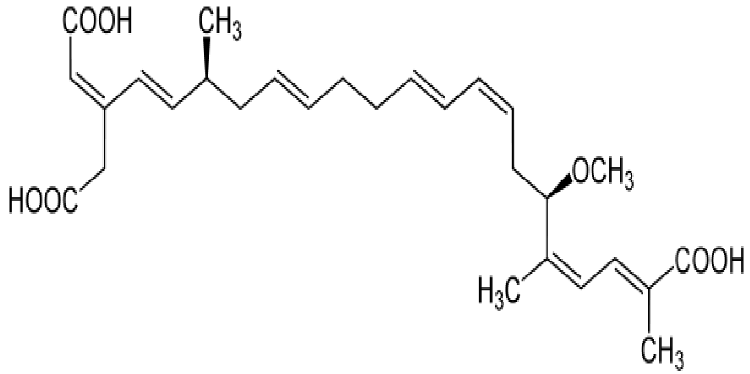
Inhibiting ANT: ADP/ATP transport

<u>Atractyloside:</u> from thistle plant *Atractylis gummifera*



Do not need to know the structures

<u>Bongkrekic acid:</u> respiratory toxin produced in coconuts contaminated with *Burkholderia gladioli*



Effect of both are similar to oligomycin

Complex V can't dissipate proton gradient if inhibited or has no substrate (ADP)

ETC will shut down as well, failure to dissipate proton gradient

SOM. MK.I.BPM2.2.DM.3.BCHM.1131

Hypoxia and ETC

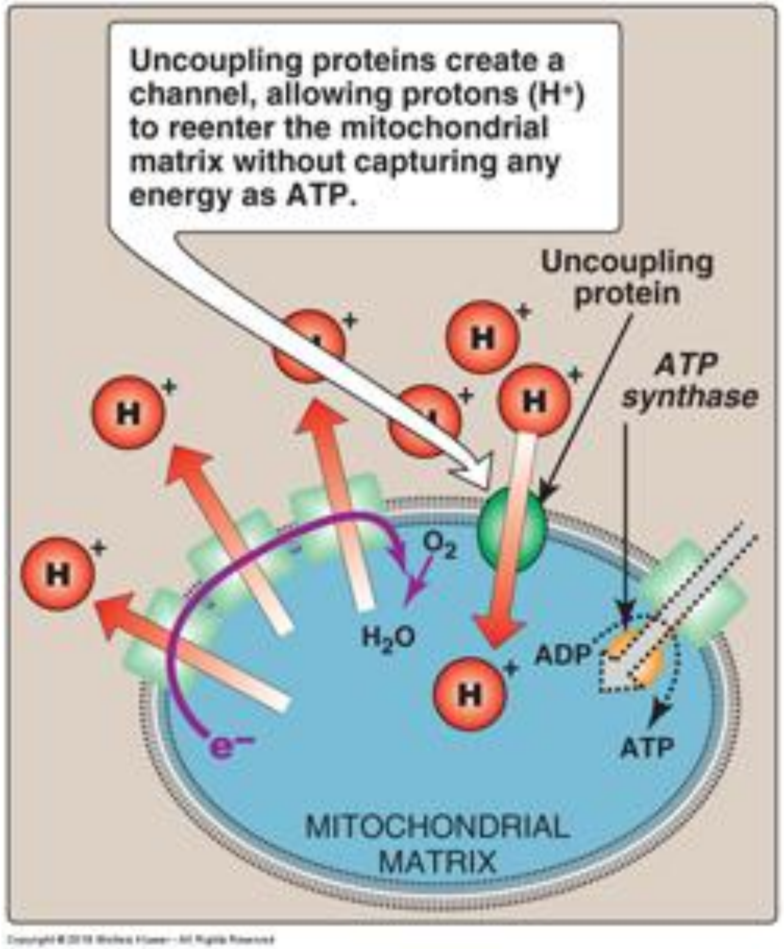
- Hypoxia decreases the rate of ETC and ATP production.

- A drop in cellular ATP increases anaerobic glycolysis and lactic acid production, anaerobic glycolysis cannot meet most tissue demands (neural tissue, cardiac muscle).

- Myocardial infarction can result from hypoxia, leading to tissue damage, leakage of intracellular enzymes (CK-MB) and troponin I and T.

Uncoupling of Oxidative Phosphorylation

- Occurs when H^+ re-enter mitochondrial matrix without going through ATP synthase
- "Dissipation" of proton gradient results in:
  - Reduction in ATP synthesis
  - Increased activity of all ETC complexes (not slowed down by proton gradient)
  - Increased Oxygen consumption (Complex IV activity increased)
  - **Release energy as heat → Hyperthermia**
- Proteins and synthetic uncouplers



Uncouplers:

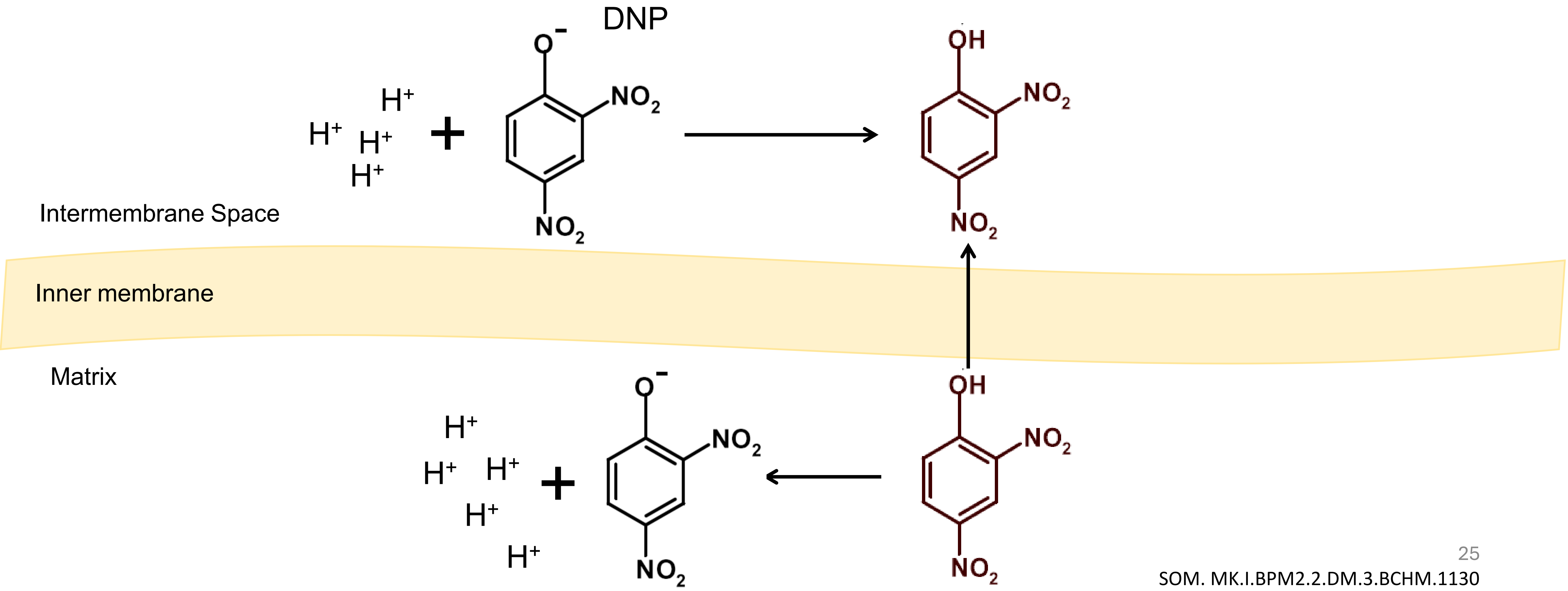
***decrease** ATP synthesis, **increase** ETC and Oxygen consumption.*

Compare ETC inhibitor to uncoupler

•SOM. MK.I.BPM2.2.DM.3.BCHM.1130

Synthetic Uncouplers

- Chemical uncouplers: Lipid-soluble
- Act by destroying the proton gradient
  - **DNP (2,4-Dinitrophenol)**
  - **ASA (aspirin)**



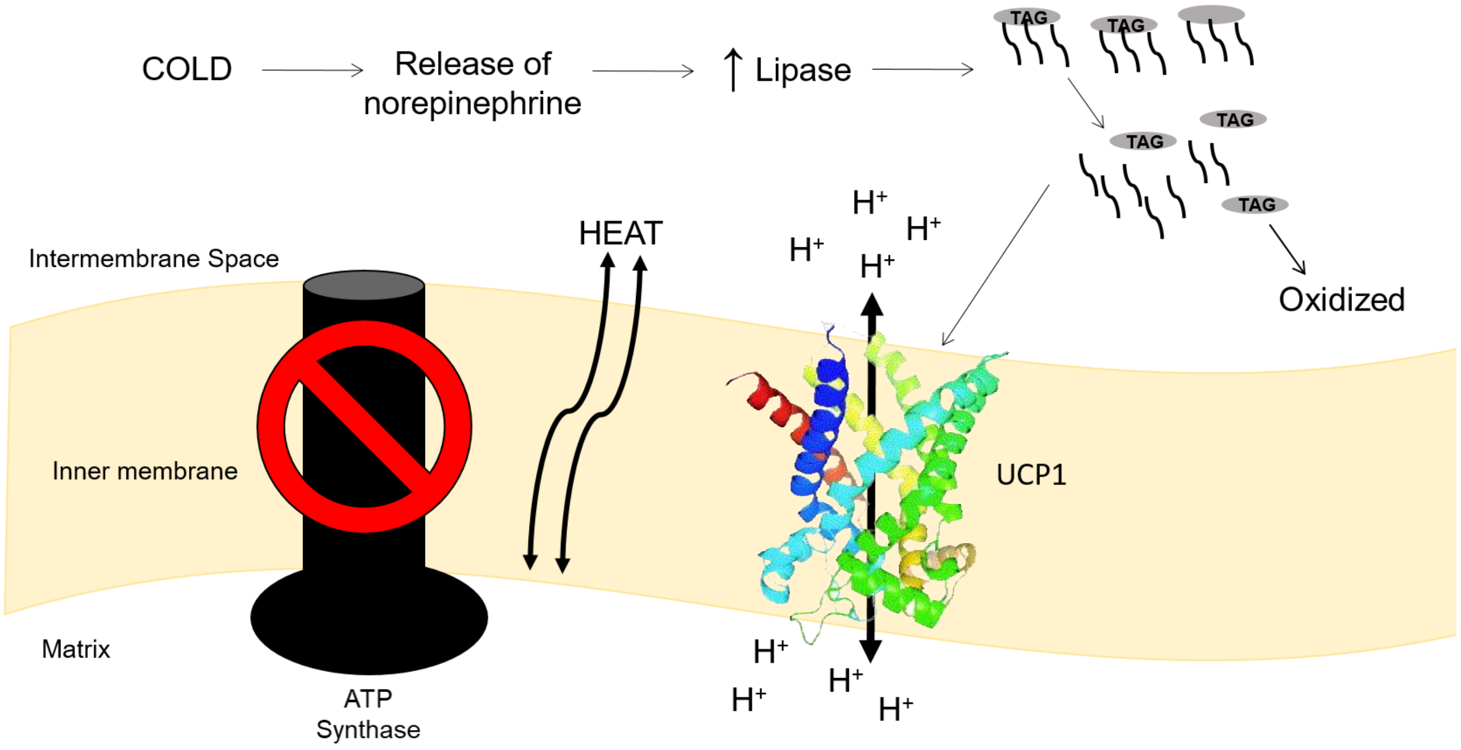
Physiological Uncouplers- Uncoupling Proteins

- Involves uncoupling proteins (UCPs)
- UCPs form channels through inner membrane which conduct H^+ back into the matrix (leaky membranes)
- When UCPs are activated, dissipation of H^+ gradient generated from electron transport is uncoupled from ATP synthesis, generates heat

UCP1- Thermogenin

- **Thermogenin** (UCP1) is expressed in **brown adipose tissue**
- Generation of heat is physiological function of brown adipose tissue
- “Non shivering” thermogenesis

Thermogenin

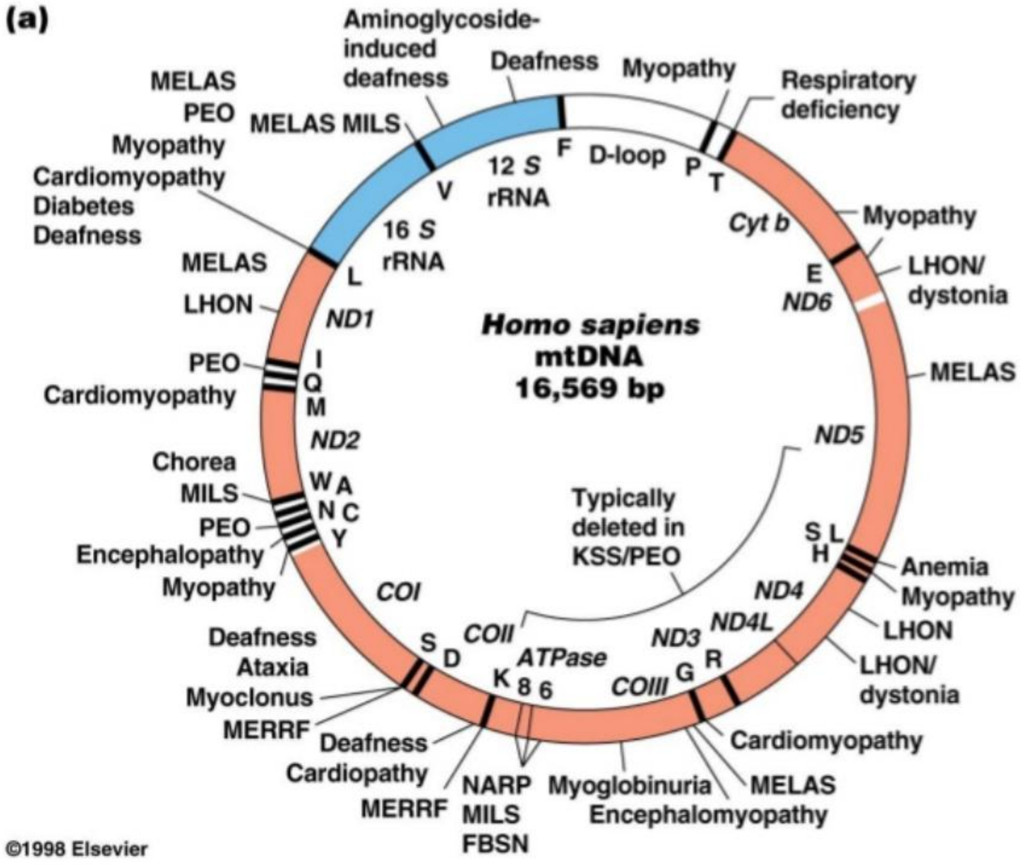


- Hormone induced release of fatty acids from triglycerides in brown fat activates thermogenin
- Thermogenin (brown) due to abundance of cytochrome-containing mitochondria
- Newborns have brown fat in neck and upper back, acts as biological heating pad
- ‘Non-shivering thermogenesis’

Mitochondrial DNA Mutations affect OxPhos

- Mitochondrial DNA (mtDNA) different from Nuclear DNA
- ETC Complexes need peptides from mtDNA and nuclear DNA
- Refer "Mitochondrial inheritance'' in Unexpected Patterns of Inheritance from Term 1

- Affects tissues with high energy demands:
  - Muscle weakness
  - Vision and/or hearing problems
  - Heart/liver/kidney problems
  - Learning disabilities



Hereditary Mitochondrial Diseases

Leber's Hereditary Optic Neuropathy

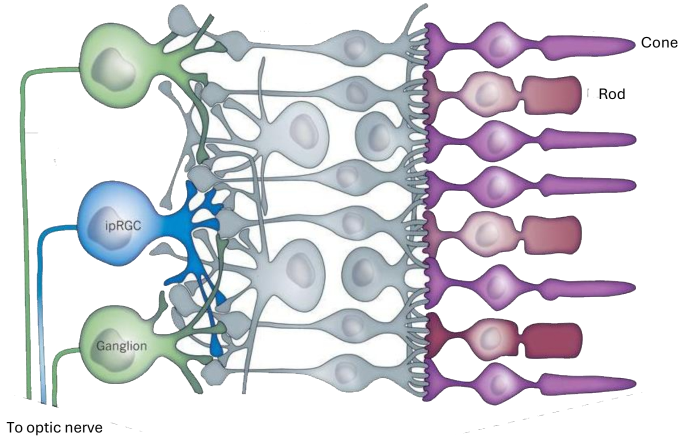
Defect in **ETC Complex I**

Deafness induced by aminoglycoside antibiotics (rRNA)

Mitochondrial myopathies (disease of the muscle):

- Kearns-Sayre syndrome (deletion in mtDNA)
- MELAS Syndrome (mitochondrial myopathy, encephalopathy, lactic acidosis, and stroke-like episodes)
- MERRF syndrome (myoclonic epilepsy; ragged red fibers)

Leber's Hereditary Optic Neuropathy



Defect in **ETC Complex I**

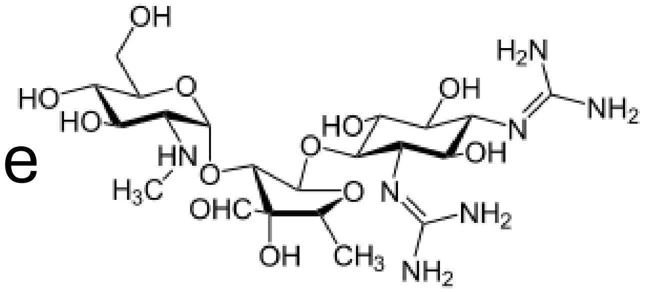
Degeneration of the retinal ganglion cells (RGCs) and atrophy of the optic nerve

Most common cause of optic atrophy

- Onset 25- 35 years (but can occur at any age) and leads to legal blindness
- Critical threshold (90%) in proportion of mutations in mtDNA must be exceeded before disease appears

Deafness induced by aminoglycoside antibiotics

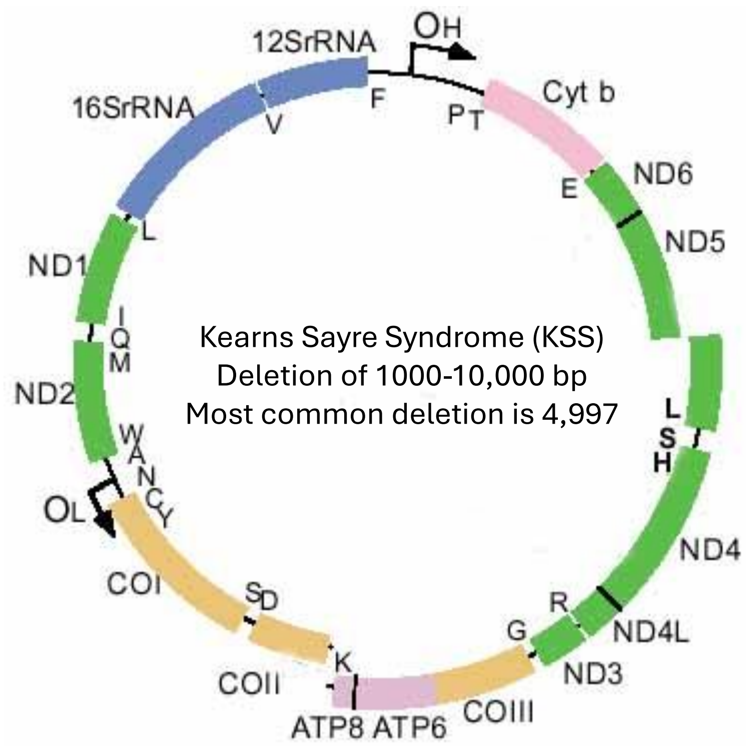
- Moderate to profound hearing loss
- Maternally inherited mutation most commonly in mitochondrially encoded **rRNA**
- Mutation may cause **predisposition** to aminoglycoside toxicity causing deafness
- Hearing loss within a few days to weeks after administration of aminoglycoside antibiotic.
- An individual with many mitochondria containing mutations may have childhood onset of hearing loss



Streptomycin
&
gentamicin

Kearns-Sayre syndrome

- Deletion within mitochondrial DNA
- Most common region is 4997 bp
- Affects systems with higher energy requirements
- Onset before 20 and includes:
  - Paralysis of eye muscles and degeneration of retina
  - Cardiac problems or congestive heart failure
  - Muscle and skeletal weakness
  - Ataxia (coordination problems)
  - Diabetes, dementia



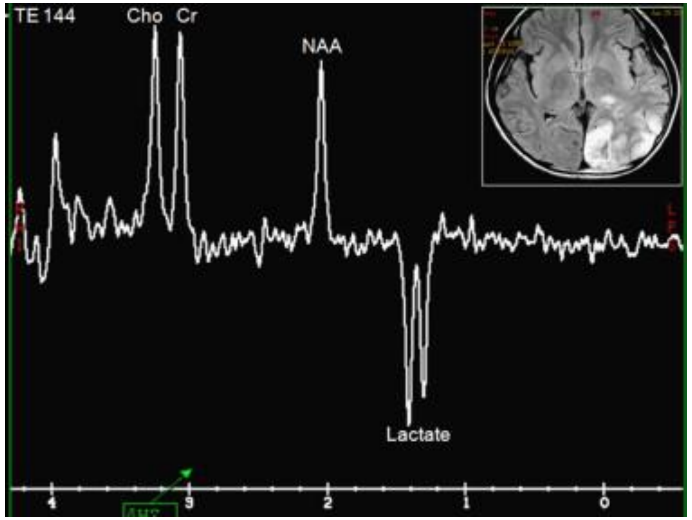
progressive external ophthalmoplegia

Shirley H. Wray, M.D., Ph.D.
Professor of Neurology, Harvard Medical School
Director, Unit for Neurovisual Disorders
Massachusetts General Hospital

SOM. MK.I.BPM2.2.DM.3.BCHM.1121

MELAS

- Mitochondrial Myopathy, Encephalopathy, Lactic Acidosis, and Stroke-like episodes
- **Most common mitochondrial disorder**
- Clinical Features: Strokes, myopathy (weak muscles), muscle twitching, dementia, and deafness
- Presentation: Stroke-like episode at 14-15 yrs of age
- Most commonly due to mutation in **mitochondrial tRNA**
- Lactic acidosis affects CNS function



Magnetic Resonance Spectra (MRS)

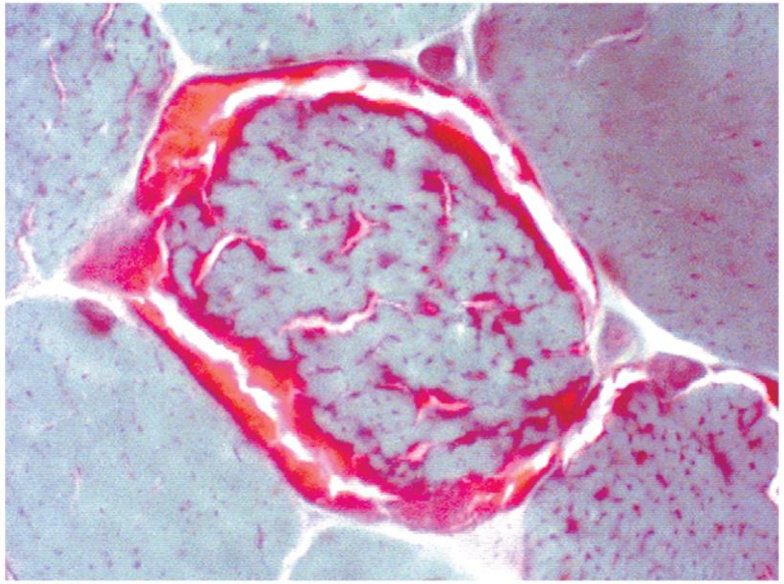
Chi et al. j.jcma, 2011

MERRF

- <u>M</u>yoclonic <u>E</u>pilepsy; <u>R</u>agged <u>R</u>ed <u>F</u>ibers
- Most common mutation in **mitochondrial tRNA gene**
- Onset: 6-16 years
- Myoclonus is usually the first symptom followed by:
  - Seizures, ataxia, muscle weakness, worsening eyesight and hearing loss

https://youtu.be/b26IsBUDo1g

(a)

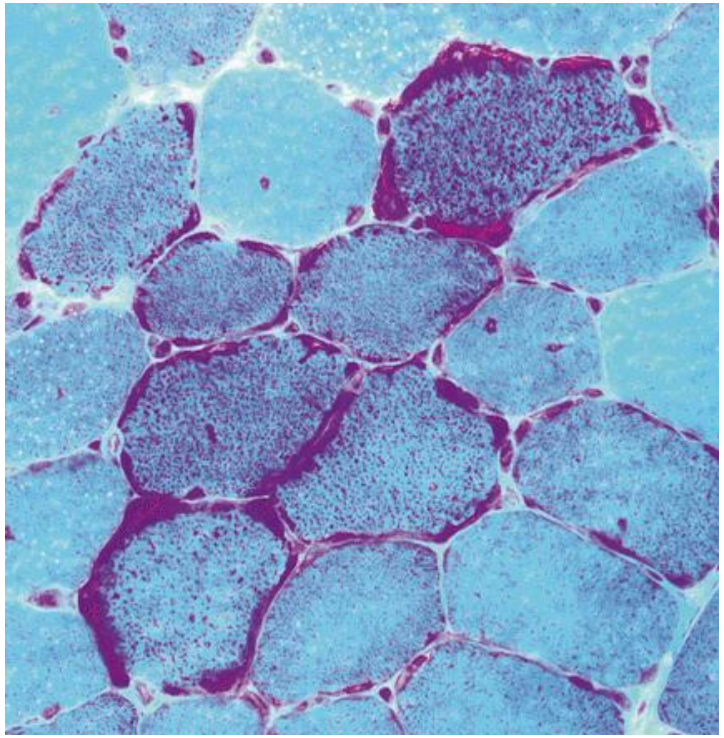


Staining of muscle cells from a patient with MERRF Ragged red fibers

SOM. MK.I.BPM2.2.DM.3.BCHM.1121

Explain “ragged red fibers”

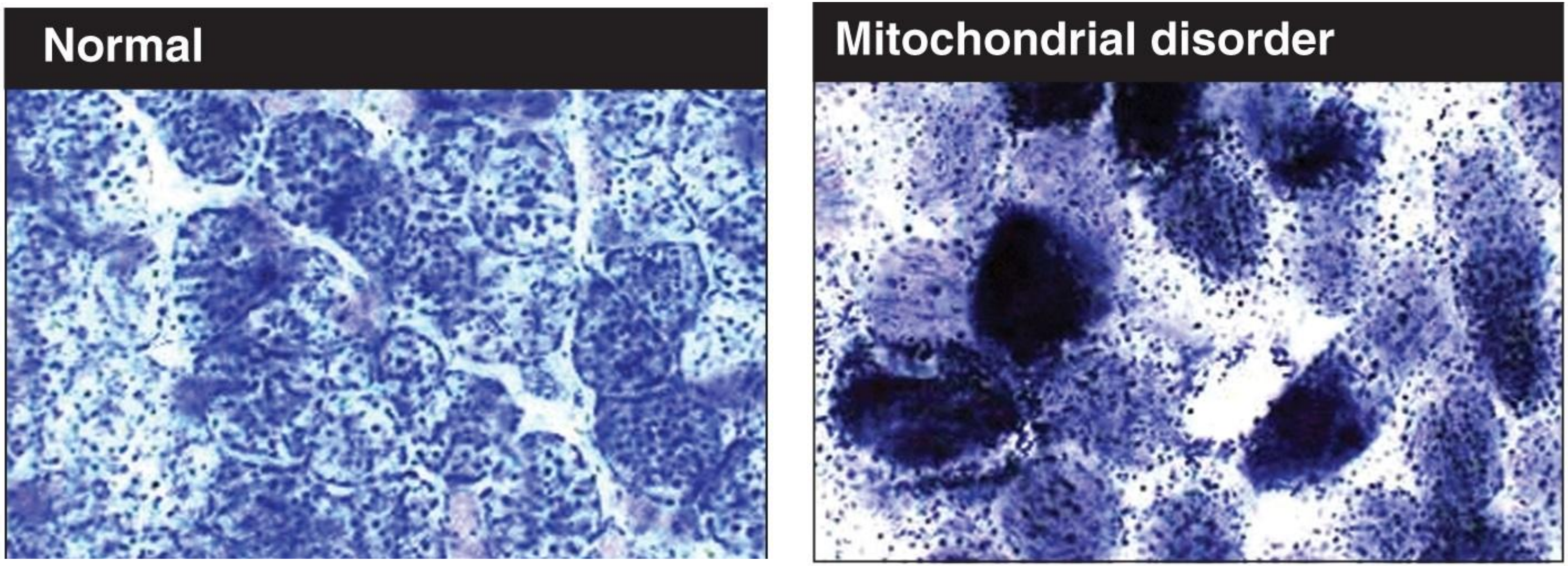
- Defective mitochondria have mutations in mtDNA or genes on nuclear DNA targeted to mitochondria
- Low amount of energy production
- The cell tries to survive low energy status by proliferating (multiplying) the mitochondria
- **Defective mitochondria proliferate and accumulate in the specific regions of muscle**
- Appearing as "Ragged Red Fibers" when muscle is stained with Gomori trichrome stain as shown



Gomori trichrome staining of a muscle biopsy from a patient with a mitochondrial disorder

Mitochondrial proliferation in muscle fibers also seen by staining for the ETC complexes

Clicker question coming up next



Stain for Complex II, Succinate Dehydrogenase

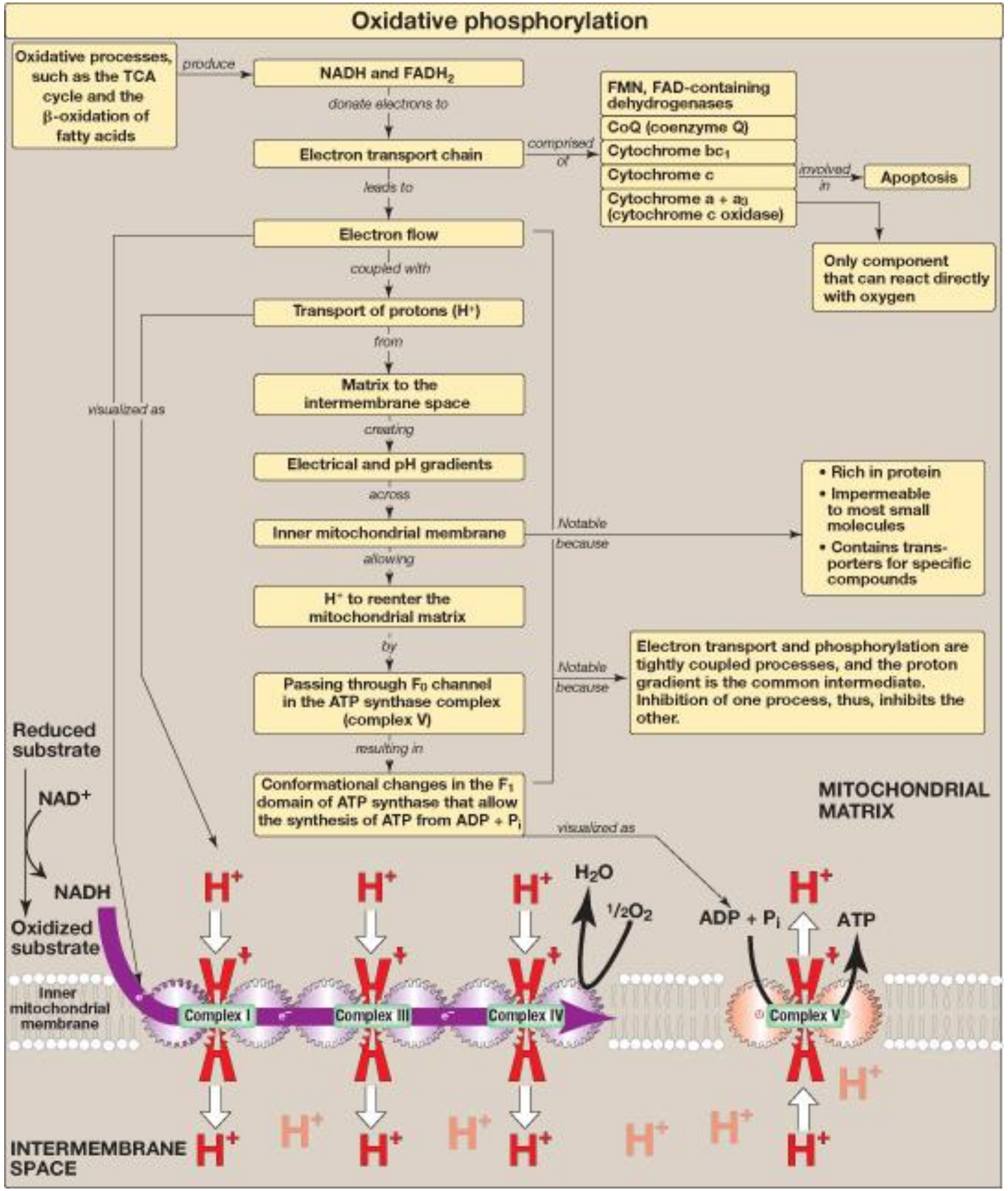
Summary Inhibitors of Oxidative Phosphorylation:

Direct Inhibitors of ETC:	**Site of Action:**	**Effect:**
Rotenone, Amytal, Piericidin A	Complex I	Shut down of ETC, no H^+ pumping, no ATP generation.
Antimycin	Complex III	
Cyanide, Azide, Hydrogen Sulfide Carbon monoxide	Complex IV	
Inhibitors of ATP Synthesis:		
Oligomycin	F_0 ATPase	Inhibition of H^+ channelling, no ATP generation, failure to dissipate H^+ gradient, shut down of ETC.
Inhibitors of ATP/ADP Transport:		
Atractyloside, Bongkrekic acid	ANT	Depletion of intra-mitochondria ADP, inhibition of ATPase, failure to dissipate H^+ gradient shut down of ETC.
Uncouplers:		
2,4-DNP, aspirin, Thermogenin (UCP)	IMM	Dissipate H^+ gradient, no ATP generation, Increase oxygen utilization; Cause hyperthermia

Compare ETC inhibitor to uncoupler

SOM.1ac.BPM2.1.DM.2. BCHM.MB.0705

Key concept map for oxidative phosphorylation (OXPHOS)



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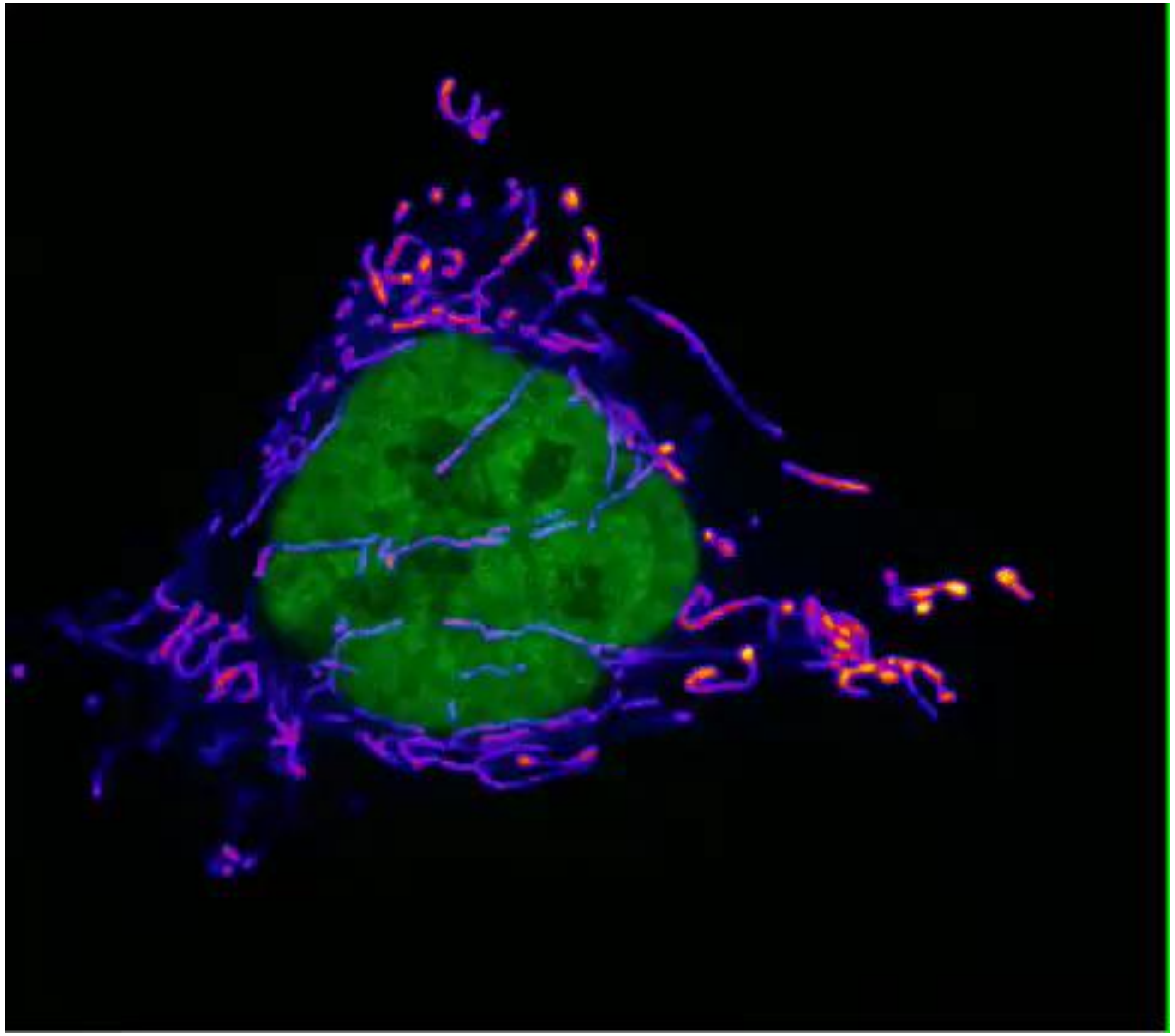

The End